

AnalysisHub

Analysis Repository Manager

User Manual

v1.0 | ANSYS Workbench 2024 R2

ACT Extension for ANSYS Workbench

1. Overview

AnalysisHub is an ANSYS Workbench ACT extension that provides a centralized file repository for every file associated with a Workbench project. It tracks reference files, supplemental databases, customer-provided data, and simulation inputs in a single structured interface — keeping your project organized, portable, and ready for review or transfer.

Feature	What it does
Centralized file tracking	All project reference files managed from a single hub window inside Workbench.
Multi-section organization	Files organized into Main WB Database, Supplemental WB Database, Customer Provided Data, and User Files sections.
Archive management	Archive files by ZIP compression or folder copy directly into the repository directory, with status tracking.
Status monitoring	Live status for every file: Ready, Archived, Source Changed, Missing, Local, and more.
Cross-machine portability	Archive paths stored relative to the repository root. Transfer the repository folder to any machine and all references resolve correctly.
Health Check	Scans for missing source files and orphaned archive files with the option to delete them.
Project information	Stores and displays project title, customer, analyst, status, and revision in a collapsible panel.

2. Requirements and Installation

2.1 Requirements

- ANSYS Workbench 2024 R2 (tested and confirmed)
- A saved Workbench project (.wbpj) — AnalysisHub requires an active project to determine the repository location.
- Write access to the project directory.

2.2 Installation

1. Copy the AnalysisHub extension folder to your ACT extensions directory, typically:

`C:\Program Files\ANSYS Inc\v242\aiso\ACT\extensions\`

2. Open ANSYS Workbench.
3. Go to **Extensions** → **Manage Extensions**.
4. Locate **AnalysisHub** in the list and click **Load**. The toolbar will appear in the Workbench ribbon.

Note: The extension only needs to be loaded once. Workbench remembers loaded extensions between sessions.

2.3 The Workbench Toolbar

After loading, two buttons appear together in the Analysis Hub group on the Workbench ribbon:

- Analysis Repository — opens the hub window for the current project.
- Help — opens this manual (PDF) from the extension's Help folder.

Note: Both buttons belong to the same toolbar group so their relationship is visually clear. The Help button also appears inside the hub window toolbar for quick access while you are working.

3. Getting Started

3.1 Save Your Project First

AnalysisHub determines the repository location from the Workbench project file. You must save the project at least once before opening the hub.

Important: Always save your project (File → Save) before clicking Analysis Repository. If no project is saved, the hub will display a message asking you to save first.

3.2 Opening the Hub

5. Save your Workbench project.
6. Click **Analysis Repository** in the Workbench ribbon.
7. The Repository Manager window opens, sized to 1360 × 900 pixels by default. It is fully resizable.

3.3 The Main Window

The Repository Manager window is organized into four areas from top to bottom:

Area	What it contains
Toolbar	Add File(s), Refresh, Open, Remove, Notes, Health Check, Archive, and Help buttons.
Project Information	Collapsible panel showing Title, Customer, Analyst, Status, and Revision fields. Click Collapse / Expand to toggle.
File Tabs	Four tabs, one per section. Each tab label shows the total file count and a X count if any files are missing.
Status Bar	Shows the current operation result and file summary at the bottom of the window.

4. Repository Sections

AnalysisHub organizes files into four sections, each on its own tab:

Section	Intended use
Main WB Database	The primary Workbench project (.wbpj or .wbpz) for which this repository is being maintained. Typically one entry.
Supplemental WB Database	Supporting ANSYS projects that provide reference data, sub-models, or boundary conditions for the main project.
Customer Provided Data	Files received from the customer: drawings, specifications, load data, geometry, reports, etc.

User Files	Point cloud data, mapping files, temperature/pressure inputs, scripts, and other files that live in the project's user_files directory.
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Note: Files in the User Files section that reside inside the project's user_files directory are automatically recognized as Local ✓ and do not need to be archived. Their paths are stored relative to the project, so they resolve correctly on any machine that receives the project.

5. Managing Files

5.1 Adding Files

8. Click the tab for the section you want to add to.
9. Click **+ Add File(s)...** in the toolbar.
10. Browse to and select one or more files. Multiple files can be selected at once.
11. Click **Open**. The files appear in the file list with a **Ready** status.

Tip: You can add files from anywhere on disk. Files inside the user_files directory will be detected automatically and flagged as Local ✓ — no archiving required.

5.2 Removing Files

12. Select the file(s) to remove in the list.
13. Click **X Remove** or right-click and choose Remove from Repository.
14. If an archived copy exists in the repository, you will be asked whether to also delete the physical archive file.

Note: Removing a file from the hub deletes only the manifest entry. The original source file is never touched.

5.3 File Status Colors

Every file in the list displays a color-coded status:

Status	Meaning
Ready	Source file found. No archive copy yet.
Local ✓	File lives inside the project's user_files directory. Already home — no archiving needed.
Archived ✓ [ZIP] or [Copy]	Archived and the archive matches the current source.
Archived X — Source Changed	Archived, but the source file has been modified since the archive was created. Re-archive recommended.
Unarchived — Archive when ready	The file was previously extracted from an archive and set as the working source, but has not been re-archived yet.
MISSING	Source file not found and no archive exists on this machine. May indicate a transfer without the source files.
△ Source Project Changed	A .wbpz file with a linked source .wbpj; the .wbpj has been modified since the .wbpz was created.
△ Source Project Missing	A .wbpz file with a linked source .wbpj; the .wbpj can no longer be found.

5.4 Notes

15. Select a file in the list.
16. Click  **Notes** or right-click and choose Edit Notes.
17. The Notes dialog opens with the filename in the title bar. Type or edit the note text.
18. Click **OK** to save. The Notes column updates immediately.

Tip: The Notes dialog is resizable. Drag the corner to expand the text area for longer notes.

6. Archiving Files

6.1 What Archiving Does

Archiving creates a preserved copy of a file (or an entire Workbench project and its associated files) inside the repository directory. The original source file is never moved or deleted. The hub tracks the archive copy and reports if the source changes after the archive was created.

6.2 Best Practice — Supplemental ANSYS Databases

For supplemental Workbench projects, the recommended approach is to reference a .wbpz archive file rather than the live .wbpj project file.

Best Practice: Before adding a supplemental database to AnalysisHub: open the project in Workbench, go to File → Archive to produce a .wbpz file, then add that .wbpz to the hub. The hub will remind you of this workflow if you attempt to add a .wbpj to the Supplemental WB Database section.

Benefits of referencing a .wbpz instead of a .wbpj:

- The .wbpz opens directly in Workbench — no extraction step needed.
- The original .wbpj is protected from accidental edits.
- Archiving the .wbpz in the hub is identical to archiving any other file.
- The supplemental database is treated as a completed reference, reflecting the intent that it provides supporting data for decisions made in the main model.

If the underlying .wbpj is later modified and a new .wbpz is produced, you can link the source project to the hub entry (see Section 6.5) so the hub detects when the .wbpz is out of date.

6.3 Archive Methods

Method	When to use it
ZIP [ZIP]	Recommended for Workbench projects (.wbpj) and any file where a compact single-file archive is preferred. Stores the project and all associated files in one .zip at compression level 1 (fast). Result opens via extraction.
Copy [Copy]	Copies the file or project folder directly into the repository directory without compression. Good for large binary files where compression provides no benefit. Supports delta re-copy (only changed files updated).

6.4 Opening the Archive Dialog

- To archive all eligible files: click the  Archive button in the toolbar.
- To archive one specific file: right-click it in the list and choose Archive this file...

The Archive dialog shows:

- All archivable files grouped by section with their current status.
- A checkbox next to each file (pre-checked for files that are Ready or have a Source Changed status).
- File size and archive date for reference.
- ZIP / Copy method selector and Include result files checkbox (for Workbench projects).

19. Review and check or uncheck files as needed.

20. Choose the archive method (ZIP recommended for .wbpj files).

21. Click **Archive Selected**. A progress dialog tracks each file.

22. When complete, the status column updates to **Archived ✓ [ZIP]** or **Archived ✓ [Copy]**.

6.5 Linking a Source Project to a .wbpz Record (Option B)

If a .wbpz file in the hub was created from a .wbpj source project that may be updated in the future, you can link the source .wbpj to the hub entry. The hub will then detect if the .wbpj has been modified since the .wbpz was created and flag the record with a  Source Project Changed status.

23. Right-click the .wbpz record in the file list.

24. Choose  **Link Source Project (.wbpj)....**

25. Browse to the originating .wbpj file and click Open.

26. The link is stored in the manifest. On every Refresh, the hub compares the .wbpj modification time to the .wbpz.

Tip: To remove the link, right-click the record and choose Clear Source Project Link. This does not affect the .wbpz file itself.

7. Opening Files

7.1 The Open Button

Select a file in the list and click ► Open (or double-click the row, or right-click → Open). The hub determines the best open mode automatically based on the file's current state.

7.2 Open Modes

Condition	What happens
Source file exists	Source opens directly. If an archive also exists, you are offered the choice of opening source or archive.
Archive exists (Copy method)	Archive copy opens directly — no extraction needed.
ZIP archive, prior extraction exists	You are offered the choice of opening the previously extracted copy, re-extracting, or opening the source (if present).
ZIP archive, no prior extraction	The ZIP is extracted to the repository directory first, then the extracted project opens.
Local ✓ file	Source opens directly from the user_files location.

7.3 Workbench Project Files (.wbpj, .wbpz)

When opening a .wbpj or .wbpz, AnalysisHub launches Workbench with the file path as the argument using the runwb2 executable. Other file types are opened with their default Windows application (Excel, Adobe Reader, Notepad++, etc.).

Note: Only one Workbench session can be open at a time. Close the current session before opening a project from the hub, or use the Open dialog's source option to open a copy in a new session.

8. Project Information Panel

The collapsible panel below the toolbar stores project-level metadata. Click ▲ Collapse or ▼ Expand to show or hide it.

Field	Purpose
Title	Descriptive name for this analysis or project.
Customer	Customer or program name.
Analyst	Name of the engineer responsible for the analysis.
Status	Current project state: Active, In Review, Complete, On Hold, or Archived.
Revision	Revision identifier (e.g., Rev A, v1.2, 2024-03-15).

Note: Project information is stored in the manifest.json file alongside the repository. It is visible to anyone who opens the hub on a machine that has received the project.

9. Health Check

Click ♥ Health Check to scan the entire repository and generate a diagnostic report. The Health Check dialog shows two sections:

9.1 Missing Source Files

Lists every manifest entry where the source file cannot be found on the current machine and no archive copy exists. This typically occurs when a project is transferred to a machine that does not have the original source files.

- Select a row and click Relink to browse to the current location of the file and update the manifest.

9.2 Orphaned Archive Files

Lists archive files found in the repository directory that are not referenced by any manifest entry. These are typically left over from removed records or failed archive operations.

- Select a row and click Delete to permanently remove the orphaned file from disk.

Note: The Health Check does not modify any manifest entries automatically. All actions (Relink, Delete) require explicit user confirmation.

10. Cross-Machine Transfer

10.1 How Paths Are Stored

AnalysisHub stores archive paths relative to the repository root directory, not as absolute paths. When the repository folder is copied or zipped and transferred to another machine, all archive paths resolve correctly regardless of where the folder is placed on the recipient machine.

10.2 What the Recipient Sees

- Files with archive copies show Archived ✓ — the archive is accessible.
- Files with Local ✓ status resolve correctly if the full Workbench project (including user_files) was transferred.
- Files where only the source path was tracked (no archive) will show MISSING until the recipient runs a Health Check and relinks them.

10.3 Recommended Transfer Workflow

27. Run ♥ **Health Check** and resolve any orphaned or missing entries before transferring.
28. Archive any files with **Ready** or **Source Changed** status that the recipient will need.
29. Use ANSYS Workbench's native **File** → **Archive** to produce a .wbpz of the main project. This captures the Workbench project, user_files, and all associated data.
30. Transfer the .wbpz to the recipient. The AnalysisHub repository directory is included in the .wbpz automatically.
31. The recipient extracts the .wbpz, opens Workbench, and launches AnalysisHub. Archived files resolve immediately.

Note: Source file paths that point outside the repository (e.g., original .wbpj locations on the sender's machine) will not resolve on the recipient's machine. This is expected — the archive copy is the portable reference.

11. Tips and Best Practices

11.1 General

- Always save the Workbench project before opening AnalysisHub. The hub derives all paths from the saved project location.
- Use one repository per project. The manifest.json lives in an AnalysisRepository subfolder inside the project's user_files directory.
- Commit the manifest.json to version control (e.g., Git) along with the project to track the history of archived files.

11.2 Supplemental Databases

- Reference .wbpz files rather than .wbpj files for completed supplemental databases. This protects the original project from accidental modification and simplifies archiving.
- If a supplemental project is still actively changing, it is acceptable to reference the .wbpj directly. Archive it to a ZIP periodically using the hub to create snapshots.
- Use the Link Source Project feature to connect a .wbpz to its originating .wbpj so the hub can warn you when the source has been updated.

11.3 Archiving

- Archive before transferring. A recipient machine without the original source files can still open every file through the hub if archives exist.
- Use ZIP for Workbench databases — it produces a single portable file and compresses efficiently.
- Use Copy for large result files or binary data where compression provides little benefit and speed matters.
- Uncheck Include result files unless the recipient specifically needs the solver results. Result files (.rst, .db) are large and rarely needed for reference purposes.

11.4 File Naming

- Use consistent, descriptive file names before adding files to the hub. The label shown in the hub is taken from the file name at the time it is added.
- Avoid spaces and special characters in file and folder names for maximum compatibility across tools.

12. Troubleshooting

Problem	Solution
Hub opens with message: no project saved	Save the Workbench project (File → Save) and then click Analysis Repository again.
File shows MISSING after transfer to another machine	The source file path does not exist on this machine. If an archive exists, the status will show Archived ✓ and you can open it. If no archive exists, run Health Check and use Relink to point to the file on this machine.
Local ✓ files show MISSING on recipient machine	The user_files directory was not included in the transfer. Ensure the full .wbpz archive (generated via File → Archive in Workbench) is used for transfer, as it includes user_files.
Archive progress dialog appears to hang	Large Workbench projects with many result files can take several minutes to compress. Check the debug log at C:\Temp\AnalysisHub_debug.log for progress details.
Status shows Archived X — Source Changed but I did not edit the file	Workbench updates session metadata (modification time) when a project is opened even without saving. AnalysisHub uses both mtime and file size to detect real changes. If the file size is unchanged, the flag is suppressed. If it still appears, re-archive to update the baseline.
⚠ Source Project Changed appears on a .wbpz record	The linked source .wbpj has been modified after the .wbpz was created. Re-archive the .wbpj to a new .wbpz using Workbench, then re-add or relink the hub entry.
Help button opens a missing file message	Place AnalysisHub_Manual.pdf in the Help\ folder inside the extension directory. The path is: <InstallDir>\Help\AnalysisHub_Manual.pdf.
Right-click Archive option is grayed out	The selected file has a Local ✓ status — it lives in the user_files directory and does not need to be archived.
Workbench project does not open after clicking Open	Confirm that runwb2.exe is accessible on this machine in the standard ANSYS installation directory. Check C:\Temp\AnalysisHub_debug.log for the exact error.

For all issues, the debug log at C:\Temp\AnalysisHub_debug.log records every operation with timestamps and error details. Include this file when reporting issues.

13. Version History

Version	Key changes
v1.0	Initial release. Centralized file repository with four sections. ZIP and Copy archive methods with delta re-archive. Status tracking (Ready, Archived, Source Changed, Missing, Local, Unarchived). Cross-machine relative archive paths. in_user_files detection with Local ✓ status. .wbpz source project linkage (Option B) with ⚠ Source Project Changed detection. Health Check with orphaned archive file cleanup. Project Information panel. Resizable Notes dialog with filename in title. Best-practice nudge for .wbpj on Supplemental tab. Help button opening PDF manual from extension directory.